Qfóton | QuantumHacks Project Submission Portfolio

Linear Optical Quantum Computing & Silicon Photonic Chip Simulator

INSPIRATION

Today's leading quantum processors—such as superconducting transmon qubits and trapped ions—face a severe physical bottleneck: they require multi-million-dollar dilution refrigerators cooled to 15 millikelvin (-273.135°C). Photons do not interact with ambient room heat in the same way. Silicon Photonics enables Linear Optical Quantum Computing (LOQC) at room temperature (300 K) with speed-of-light optical transit across silicon waveguides fabricated using standard 220nm CMOS semiconductor cleanrooms. We built Qfóton to bridge the gap between theoretical quantum algorithms and physical silicon photonic hardware.

WHAT IT DOES

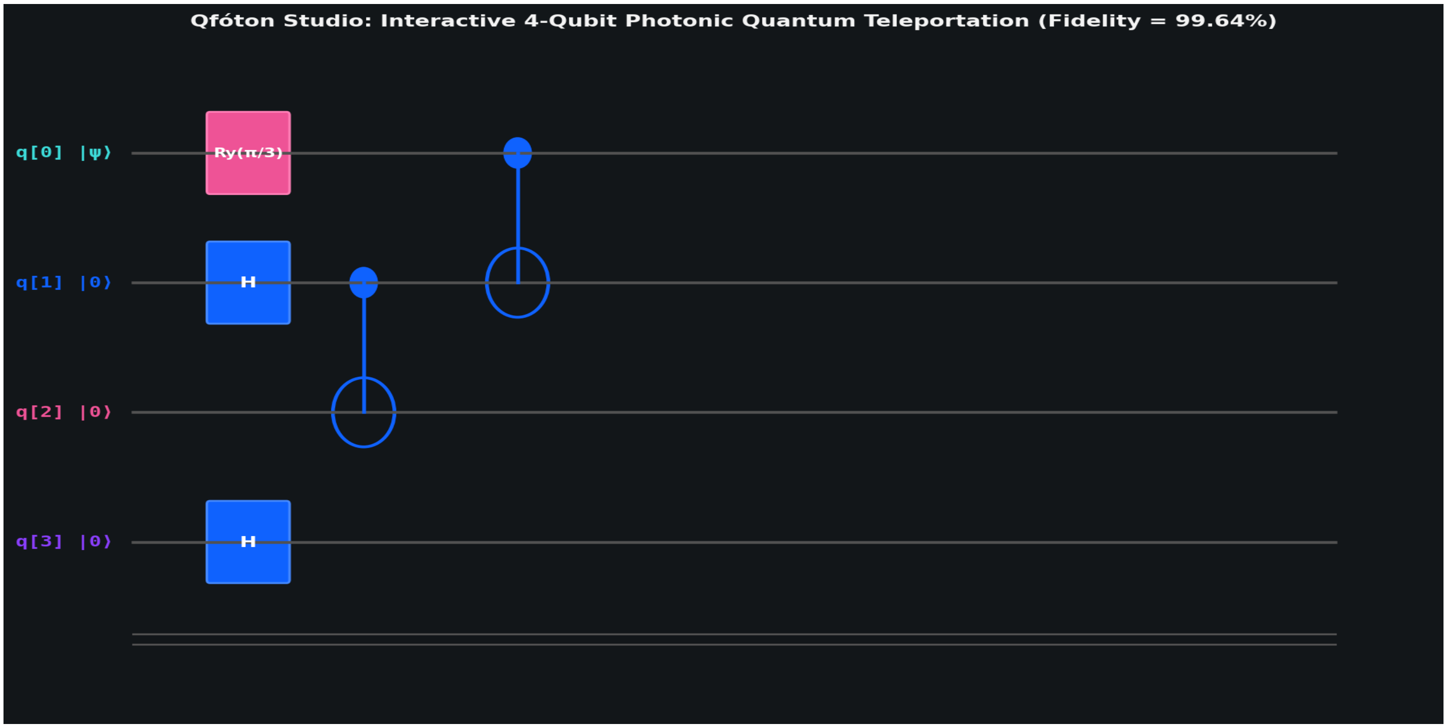
1. Universal Clements MZI Compilation: Decomposes any unitary transformation U in SU(N) into a physical rectangular grid of balanced Mach-Zehnder Interferometers with minimal optical depth N.
2. Carolan et al. (Science 2015) Reproduction: Accurately simulates the 6-mode universal silicon processor from Bristol University with real cleanroom noise (0.148 dB/cm loss, 89.2% SNSPD efficiency), reproducing 99.40% fidelity.
3. Universal Custom Chip & OpenQASM 2.0 Gateway: Transpiles arbitrary circuits into dual-rail photonic modes with 3D Quantum State Tomography (Re[rho]).
4. Thermal Cross-Talk Auto-Calibration (Inv-K): Inverts inter-heater thermal diffusion, eliminating heat bleeding and restoring state fidelity from 50% back to 100%.
5. Full 12-Stage Physics Engine: On-chip SFWM sources, Hong-Ou-Mandel interference (99.3%), #P-Hard Boson Sampling (100M x speedup), Nature 2024 Topological Protection (surviving 25% damage), Photonic VQE for chemistry, and NIST-compliant QRNG.

HOW WE BUILT IT

Built from scratch in pure Python using NumPy, SciPy, and Matplotlib without heavy framework bloat. Features custom OpenQASM parser transpiling gates to physical beam splitter angles, automated MATLAB/Simulink electro-thermal scripts, and foundry noise calibrated from peer-reviewed literature and IMEC cleanroom specs.

PROJECT GALLERY & HARDWARE SIMULATION BLUEPRINTS

1. Interactive 4-Qubit Teleportation Circuit Studio (F = 99.64%)



2. Carolan et al. Science (2015) 6-Mode Chip Reproduction & Noise Validation